Use Case - Uncertainty Quantification for R^2 in Multiple Linear Regression

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Background & Literature (tbc)

Research Goal

This use case will quantify uncertainty in the coefficient of determination, R^2, by treating it as a function of the joint correlation matrix R:

$$R^2 = R_{XY}^T R_{XX}^{-1} R_{XY}$$

We will propagate uncertainty in the estimated joint correlation matrix $\hat{R}$ to uncertainty in $R^2(\hat{R})$, and evaluate how this uncertainty changes across sample size, signal strength, and conditioning of the predictor correlation structure R_{XX}.

Methods

We will study uncertainty in the coefficient of determination, R^2, by treating it as a function of the joint correlation matrix of (Y, X). For standardized variables, the population R^2 can be written as

$$R^2(R) = R_{XY}^T R_{XX}^{-1} R_{XY},$$

where R_{XX} is the predictor correlation matrix and R_{XY} is the vector of correlations between the predictors and the outcome. This formula allows uncertainty in the estimated joint correlation matrix $\hat{R}$ to be propagated directly to uncertainty in R^2.

For each simulation scenario, we will first specify a population predictor correlation matrix R_{XX}, a coefficient vector β, and a target population value R^2_{true}. The data-generating model is

$$Y = X^T\beta + \varepsilon, \qquad X \sim N(0, R_{XX}), \qquad \varepsilon \sim N(0, \sigma^2),$$

with $X \perp \varepsilon$. Given R_{XX} and β, the variance of signal ($X^T\beta$) is

$$S = \beta^T R_{XX} \beta.$$

The error variance σ^2 will be chosen to achieve the target population R^2:

$$\sigma^2 = S\left(\frac{1 - R^2_{\text{true}}}{R^2_{\text{true}}}\right).$$

This design allows scenarios with different predictor correlation structures and signal patterns to be compared at the same overall level of explanatory strength.

For each generated dataset, we will estimate the empirical joint correlation matrix

$$\hat{R} = \mathrm{Cor}(Y, X),$$

with block structure

$$\hat{R} = \begin{pmatrix} \hat{R}_{YY} & \hat{R}_{YX} \\ \hat{R}_{XY} & \hat{R}_{XX} \end{pmatrix}.$$

The observed coefficient of determination will be computed as

$$\hat{R}^2 = \hat{R}_{XY}^T \hat{R}_{XX}^{-1} \hat{R}_{XY}.$$

To quantify uncertainty, we will generate a 95% reverse-containment perturbation cloud of plausible joint correlation matrices around $\hat{R}$ using our MCMC approach:

$$\left\{ \hat{R}^{(b)} \right\}_{b=1}^{B}, \qquad B = 20{,}000.$$

For each perturbed matrix, we will compute

$$R^{2(b)} = \hat{R}_{XY}^{(b)T} \left(\hat{R}_{XX}^{(b)} \right)^{-1} \hat{R}_{XY}^{(b)}.$$

We will summarize this distribution of R^2 using mean, median, min, max, and interval width. Histograms will also be used to visualize the distribution of R^2.

Calibration

We will repeat M times the above simulation (set $M = 200$). Across repeated simulation replicates, we will evaluate calibration by comparing the nominal 95% perturbation intervals with the known population value R^2_{true}. Empirical coverage will be computed as

$$\text{Empirical coverage} = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1} \left\{ R^2_{\text{true}} \in [L_m, U_m] \right\},$$

where $[L_m, U_m]$ is the min and max of R^2 from replicate m. The mean interval width will be computed as

$$\text{Mean interval width} = \frac{1}{M} \sum_{m=1}^{M} (U_m - L_m).$$

Simulation Design: Specifying R_{XX}, β, Target R^2, and n

The simulation scenarios will vary the predictor correlation structure R_{XX}, coefficient vector β, and the target population R^2. These factors determine the population joint correlation structure and therefore the corresponding mapping to $R^2 = R_{XY}^T R_{XX}^{-1} R_{XX}$

The predictor correlation matrix R_{XX} determines the geometry of the predictor space. We consider an AR(1) structure,

$$(R_{XX})_{ij} = \rho^{|i-j|},$$

where $\rho \in \{0.2, 0.5, 0.8\}$, representing weak, moderate, and strong predictor correlation. Larger values of ρ induce stronger colinearity among nearby predictors, making R_{XX} more ill-conditioned, leading R_{XX}^{-1} to be more sensitive to perturbations.

The coefficient vector β determines how the signal is distributed across the predictor space. Different β scenarios will be evaluated since R^2 is determined by the signal quantity $\beta^T R_{XX} \beta$, not only on the magnitude of individual coefficients. When predictors are correlated, the same coefficient size can contribute differently to R^2 depending on whether the signal is concentrated among highly correlated predictors, spread (evenly) across predictors, or having flipped signs.

To compare scenarios at the same level of overall explanatory strength, we will specify target values of R^2_{true} and choose σ^2 accordingly. To account for sampling variability, sample size of $n \in (200, 500, 1000)$ will be evaluated in this simulation.

Table 1: Specification of different signal scenarios

Signal scenario	Signal pattern	Example β structure	Purpose
S1	All strong signals	$\beta_j = 0.6$ for all signal predictors	Evaluates uncertainty in R^2 when the signal is strong and distributed across predictors
S2	All weak signals	$\beta_j = 0.2$ for all signal predictors	Evaluates whether weak distributed signals lead to wider uncertainty in R^2
S3	Mixed weak and strong signals	Some $\beta_j = 0.6$, others $\beta_j = 0.2$	Evaluates how heterogeneous signal strength affects uncertainty in R^2
S4	One strong signal, otherwise weak	One $\beta_j = 0.8$, others $\beta_j = 0.2$	Evaluates whether a dominant predictor stabilizes R^2, especially under correlated predictors
S5	Positive/negative sign flip	β_j alternates between $+0.4$ and -0.4	Evaluates cancellation effects and whether opposing signals increase uncertainty in R^2

Simulation Procedure

For each combination of ρ, signal pattern, target R^2_{true}, and sample size n, we will:

1. construct R_{XX} using the AR(1) structure $(R_{XX})_{ij} = \rho^{|i-j|}$
2. specify β according to the signal scenario
3. compute $S = \beta^T R_{XX} \beta$
4. choose $\sigma^2 = S(1 - R^2_{\text{true}})/R^2_{\text{true}}$
5. generate $X \sim N(0, R_{XX})$ with sample size n
6. generate $Y = X^T \beta + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$
7. estimate $\hat{R} = \text{Cor}(Y, X)$
8. map $R \mapsto R^2$ by computing $\hat{R}^2 = \hat{R}_{XY}^T \hat{R}_{XX}^{-1} \hat{R}_{XY}$
9. generate a 95% reverse-containment perturbation cloud around $\hat{R}$;

10. compute $R^{2(b)}$ for each perturbed matrix;
11. summarize the distribution of perturbed R^2 and evaluate calibration across repeated simulation replicates.

Three analysis will be carried out:

- Primary analysis: we will use the 95% reverse-containment perturbation cloud and summarize uncertainty of R^2.
- Secondary analysis 1: bootstrap-based uncertainty

  This analysis will be served as a benchmark against our perturbation approach. Step 9 above will be replaced by bootstrapping data (step 6) instead of the MCMC approach. The resulting R^2 will be reported using the 2.5th and 97.5th percentiles.
- Secondary analysis 2: train/ test split

  This analysis aims to measure out-of-sample predictive performance (and its uncertainty).

  - In step 6, we will split the full dataset (Y, X) into training and test sets to obtain $\hat{R}_{train}$ and $\hat{R}_{test}$.
  - In step 9, $\hat{R}_{train}$ will be used to generate 95% reverse-containment perturbation cloud.
  - In step 10, instead of $R^{2(b)}$, we will compute the out-of-sample performance $R^{2(b)}_{test}$ using

  $$R^{2(b)}_{\text{test}} = 2\hat{\beta}^{(b)T}_{\text{train}}\hat{R}_{XY,\text{test}} - \hat{\beta}^{(b)T}_{\text{train}}\hat{R}_{XX,\text{test}}\hat{\beta}^{(b)}_{\text{train}}$$

  where

  $$\hat{\beta}^{(b)}_{\text{train}} = \left(\hat{R}^{(b)}_{XX,\text{train}}\right)^{-1}\hat{R}^{(b)}_{XY,\text{train}}$$

Results & Table Shells

Table 2: (Primary Analysis) R^2 uncertainty summary from one simulated dataset

Signal pattern	ρ	Sample Size n	True R^2	Observed $\hat{R}^2$	Perturbation median	Perturbation mean	Min L	Max U	Range
S1: All strong	0.2	200							
S2: All weak	0.2	200							
S3: Mixed weak/strong	0.2	200							
S4: One strong, otherwise weak	0.2	200							
S5: Sign flip	0.2	200							
S1	0.5	200							
S2	0.5	200							

Table 3: (Primary Analysis) Calibration summary for R^2's uncertainty across repeated simulation (simulate 200 times).

Signal pattern	ρ	Sample Size n	True R^2	Mean observed $\hat{R}^2$	Bias	MonteCarlo SD of observed $\hat{R}^2$	Empirical Coverage	Mean interval width
S1: All strong	0.2	200						
S2: All weak	0.2	200						
S3: Mixed weak/strong	0.2	200						
S4: One strong, otherwise weak	0.2	200						
S5: Sign flip	0.2	200						
S1	0.5	200						
S2	0.5	200						

Tables 4 & 5: (Secondary Analysis 1 - Bootstrap), same tables as Tables 2 & 3.

Table 6: (Secondary Analysis 2 - OOS performance), one simulation

Signal pattern	ρ	n_{train}	n_{test}	True R^2	Training $\hat{R}^2$	Test R^2_{test}	95% CI (R^2_{test})

Table 7: (Secondary Analysis 2 - OOS performance), calibration

Appendix

Linear Additive Model Consider the linear additive model setup:

$$Y = X^T\beta + \varepsilon$$

where $X \sim N(0, \Sigma_{XX})$, $\varepsilon \sim N(0, \sigma^2)$ and $X \perp \varepsilon$.

Here, X is a $p \times 1$ vector of predictors, Σ_{XX} is the predictor correlation matrix, and β is a $p \times 1$ regression coefficient vector. True signal variables are defined by nonzero regression coefficients, while null/noise variables have coefficients equal to zero.

Population Joint Correlation/ Covariance Structure The predictor correlation structure R_{XX} is specified first. Given R_{XX}, β, and σ^2, the population joint covariance matrix of (Y,X) is

$$\Sigma = \begin{bmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{bmatrix}$$

where

$$\begin{aligned}
\Sigma_{YY} &= Var(Y) \\
&= Var(X^T\beta + \varepsilon) \\
&= \beta^T \Sigma_{XX} \beta \ + \ \sigma^2 \\
\Sigma_{XY} &= Cov(X, Y) \\
&= Cov(X, X^T\beta + \varepsilon) \\
&= \Sigma_{XX}\beta
\end{aligned}$$

If predictors are standardized, then $\Sigma_{XX} = R_{XX}$. The joint covariance matrix can then be converted to a joint correlation matrix. The MCMC perturbation procedure operates in this correlation space.

Mean signal stability: Among the true signal variables in this block, what is the average stability score? How often are they selected on average?

Average #signals selected: Across perturbation iterations, how many true signal variables from this block were selected on average

AR(1) Correlation Structure For example, a R_{XX} structure with $p = 5$ is as follows:

$$R_{XX} = \begin{pmatrix} 1 & \rho & \rho^2 & \rho^3 & \rho^4 \\ \rho & 1 & \rho & \rho^2 & \rho^3 \\ \rho^2 & \rho & 1 & \rho & \rho^2 \\ \rho^3 & \rho^2 & \rho & 1 & \rho \\ \rho^4 & \rho^3 & \rho^2 & \rho & 1 \end{pmatrix},$$